

Think-Up

Product & Engineering Case Study

A Collaborative Study Platform

Study Smarter, Together.

The first AI-powered study platform built specifically for the Saudi academic journey — where students across different majors who share the same courses find and join live study sessions.

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At a glance. A working, multi-page web application backed by a real-time cloud database, serverless functions, live video, and an AI summarisation pipeline — designed and built end to end. Live demo, source, and design system referenced throughout.

1 Executive Summary

Think-Up is a collaborative study platform for Saudi university students. Its defining idea is simple but structural: study sessions are attached to a *course*, not to a *major*. A computer-science student and a software-engineering student who both take the same calculus course can discover each other, join the same live room, study together, and walk away with an AI-generated summary of what was covered.

The project was built as a course submission for *CCSW321 — Web Development* at the University of Jeddah, but it was engineered as a real product: real authentication, a real-time database, serverless backend validation, live video with transcription, and an AI summarisation layer.

Multi-page web app

Real-time database

Serverless backend

Live video + transcription

AI summaries

Bilingual (AR/EN)

THE CORE INNOVATION

Conventional study tools group students by their **major**. Think-Up groups them by the **course code** they actually share. This single architectural decision — sessions reference `courseCode`, never `majorId` — is what unlocks genuine cross-major collaboration and is the foundation the entire product is built on.

Quick facts

Product	A collaborative, AI-assisted study platform for Saudi students
Role	Full-stack lead — design, frontend, backend, infrastructure
Frontend	Vanilla HTML5, CSS3, JavaScript (no framework)
Backend	Firebase — Firestore, Auth, Cloud Functions (Node 22, 2nd Gen)
Live sessions	Daily.co video + Deepgram transcription
Intelligence	Claude API for session summaries
Pages	15 consolidated, fully bilingual English / Arabic screens
Catalogue	278 courses across 6 specialisations

2 The Problem & The Insight

2.1 What students actually struggle with

University study is fragmented. Students sit in the same lecture halls for shared, cross-faculty courses — mathematics, programming fundamentals, physics, languages — yet the tools they use to organise study isolate them inside their own department. Finding

the right person to revise a specific course with, at a specific time, is left to luck and group chats.

2.2 The reframing

The breakthrough was to stop modelling the world around *who you are* (your major) and model it around *what you are studying* (the course). A shared course code is a far better coordination key than a shared department, because it maps directly onto the moment a student is actually trying to solve: *“I need to revise this exact course, now, with someone who is doing the same.”*

Why this matters technically. Because sessions are keyed on `courseCode`, the same live room is reachable by students from any major whose curriculum includes that course. Discovery, access control, and the weekly schedule all fall out of this one relationship. Break it, and the product stops being cross-major.

2.3 Who it is for

Saudi university students who share common courses across specialisations, with a first deployment targeting the University of Jeddah’s College of Computer Science and Engineering and its six majors.

3 Product Walkthrough

This section walks through the live product screen by screen, following the path a real student takes: arrive, orient, discover a course, find or create a session, study live, and learn who built it.

3.1 Landing — the first impression

The entry point states the promise plainly and routes the visitor into the two things they can do first: explore the network of universities, or get started. A four-step “Path to Excellence” explains the model — find your university, select your major, choose your course, receive an AI summary — and a dedicated band introduces the AI note-taking value proposition.

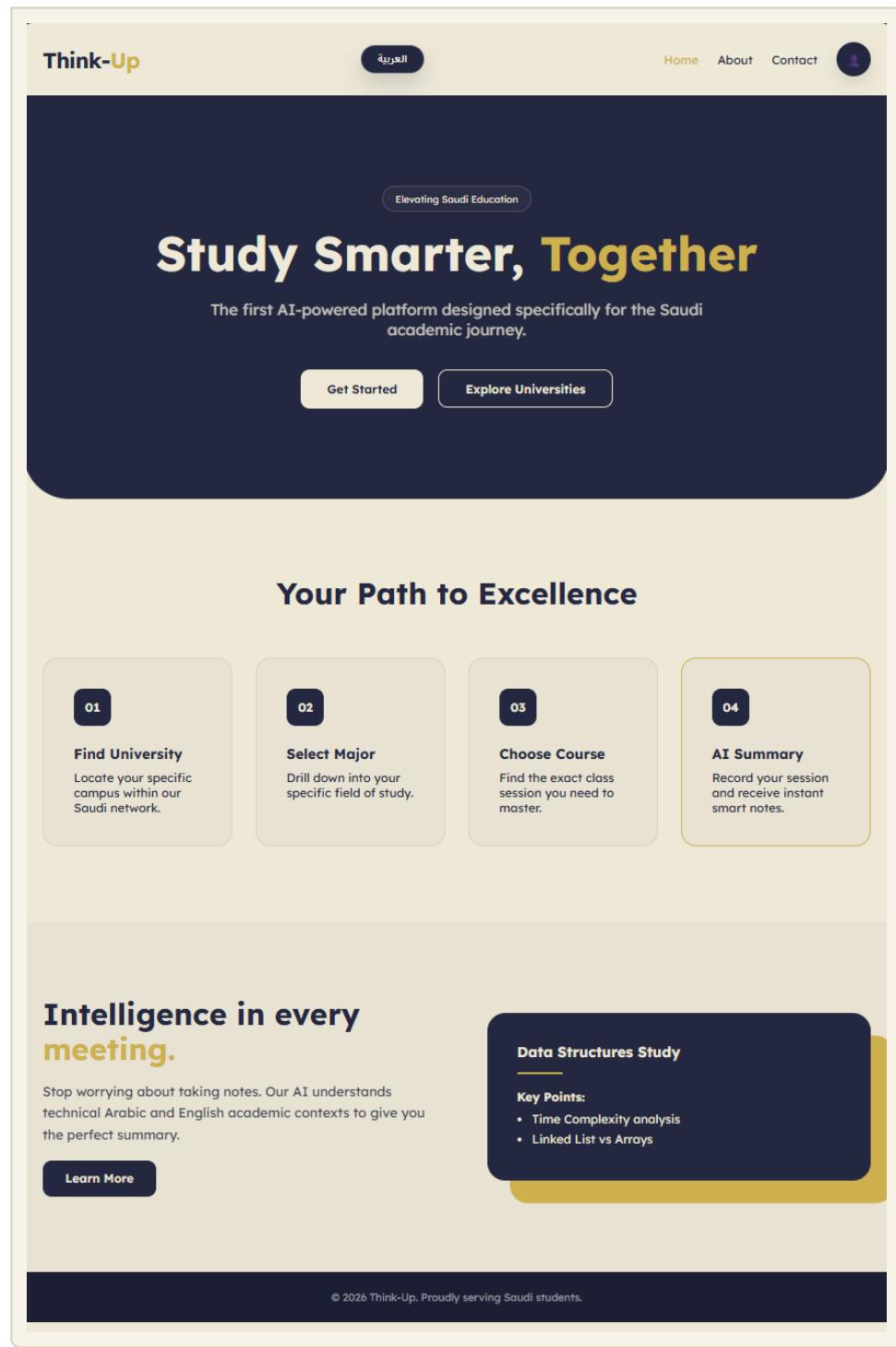


Figure 1 – The landing page: a clear value proposition, the four-step student journey, and the AI-summary pitch, all in the brand's indigo / cream / gold system.

3.2 Dashboard — the personal hub

After signing in, the student lands on a personalised dashboard. A “Today at a glance” strip surfaces upcoming sessions and the number of saved summaries; Quick Actions shorten the path to the most common next step; and a Recent Activity feed lists completed sessions. Upcoming sessions and saved AI summaries each get their own entry point.

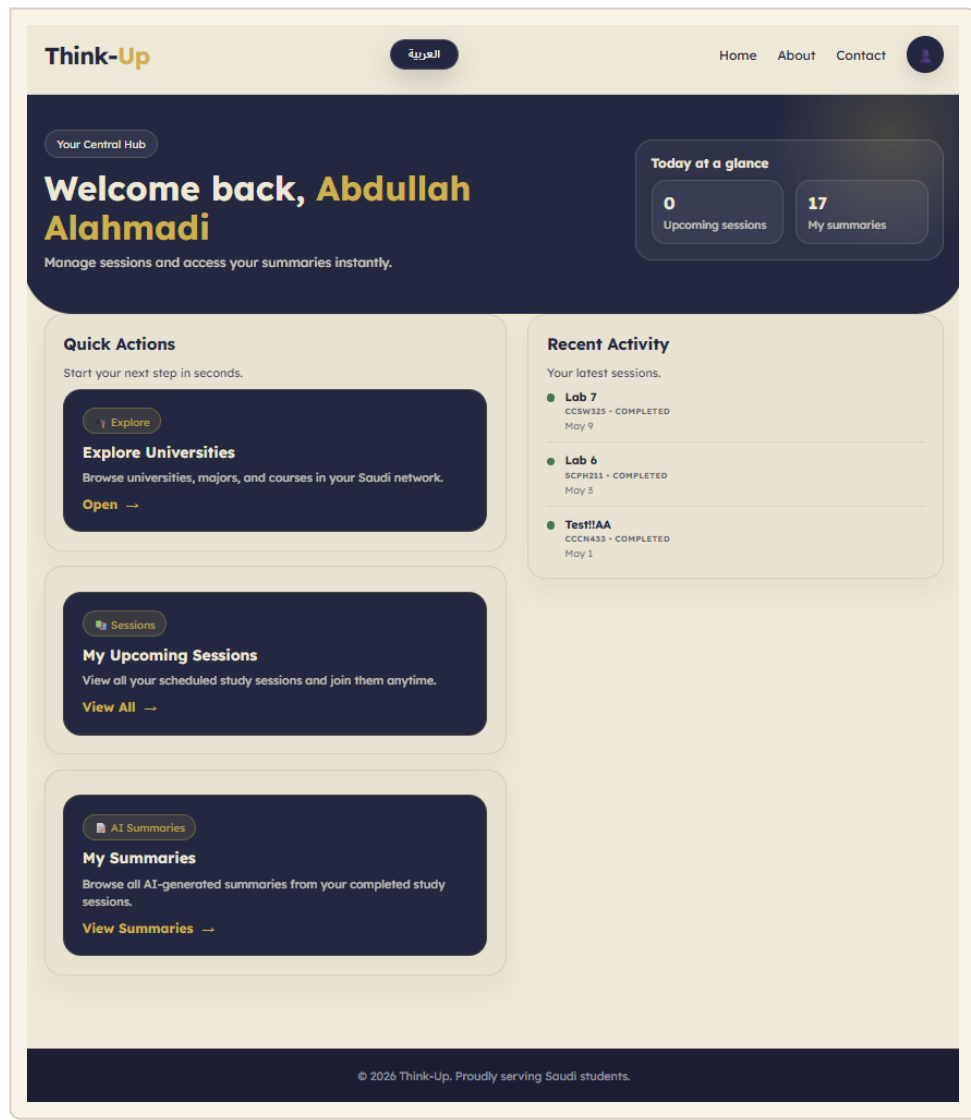


Figure 2 – The dashboard: live counts, quick actions, recent activity, and direct access to upcoming sessions and AI summaries.

3.3 Discovery — from university to course

Discovery is a deliberate funnel that mirrors the real academic hierarchy: **university** → **college** → **major** → **course**. Each step is searchable, and every course shown is backed by a real catalogue of 278 courses with curriculum references across six specialisations.

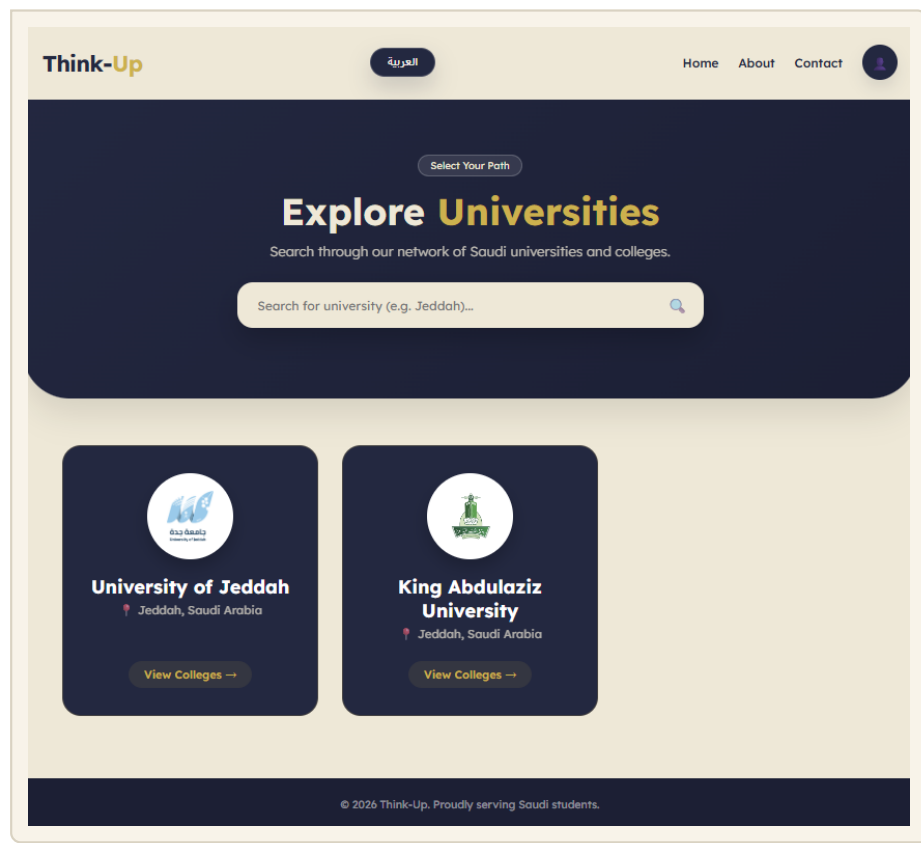


Figure 3 – Step 1 — Explore Universities. A searchable network of Saudi universities, each rendered as a card that leads into its colleges.

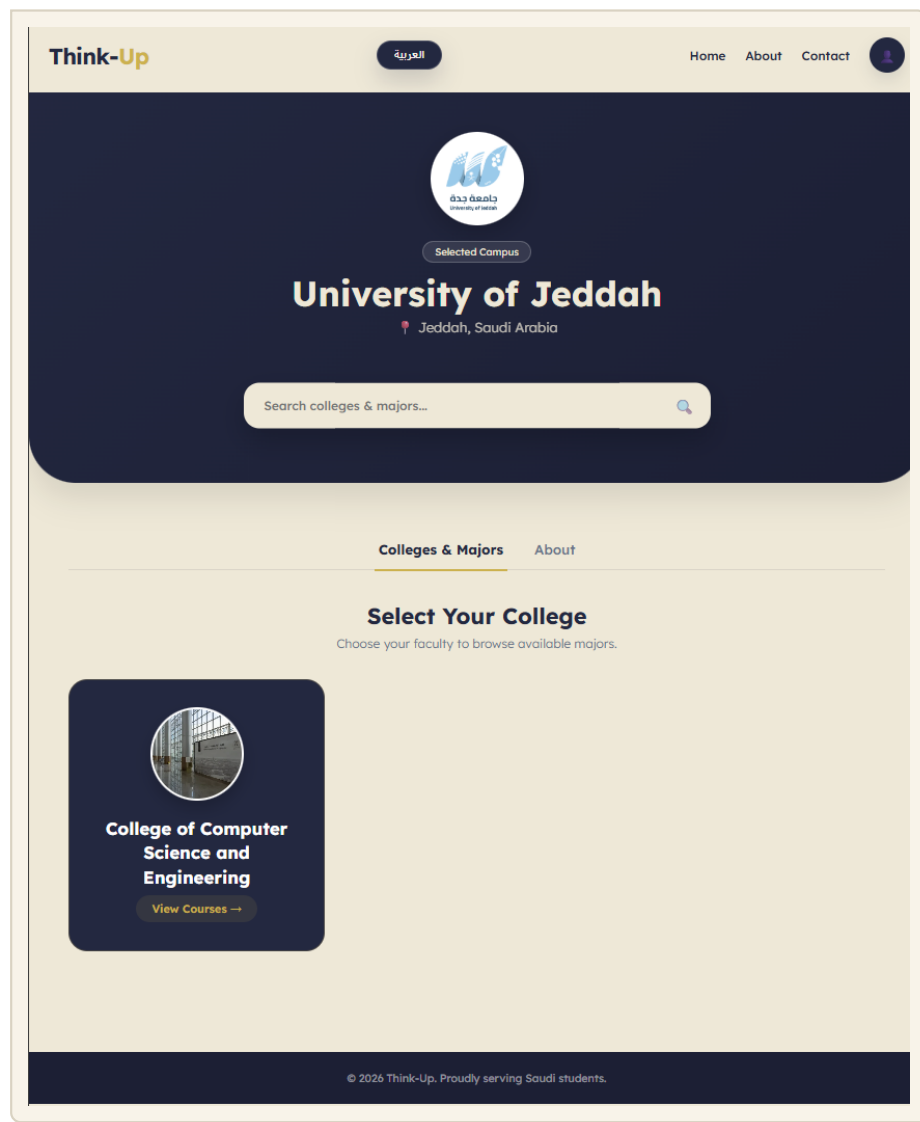


Figure 4 – Step 2 — University view. Inside a university, the student selects a college to browse its available majors.



Figure 5 – Step 3 — Choose your specialisation. The college reveals its majors — Software Engineering, Artificial Intelligence, Computer Engineering, Computer Science, Cybersecurity, Data Science — each with its credit-hour load.

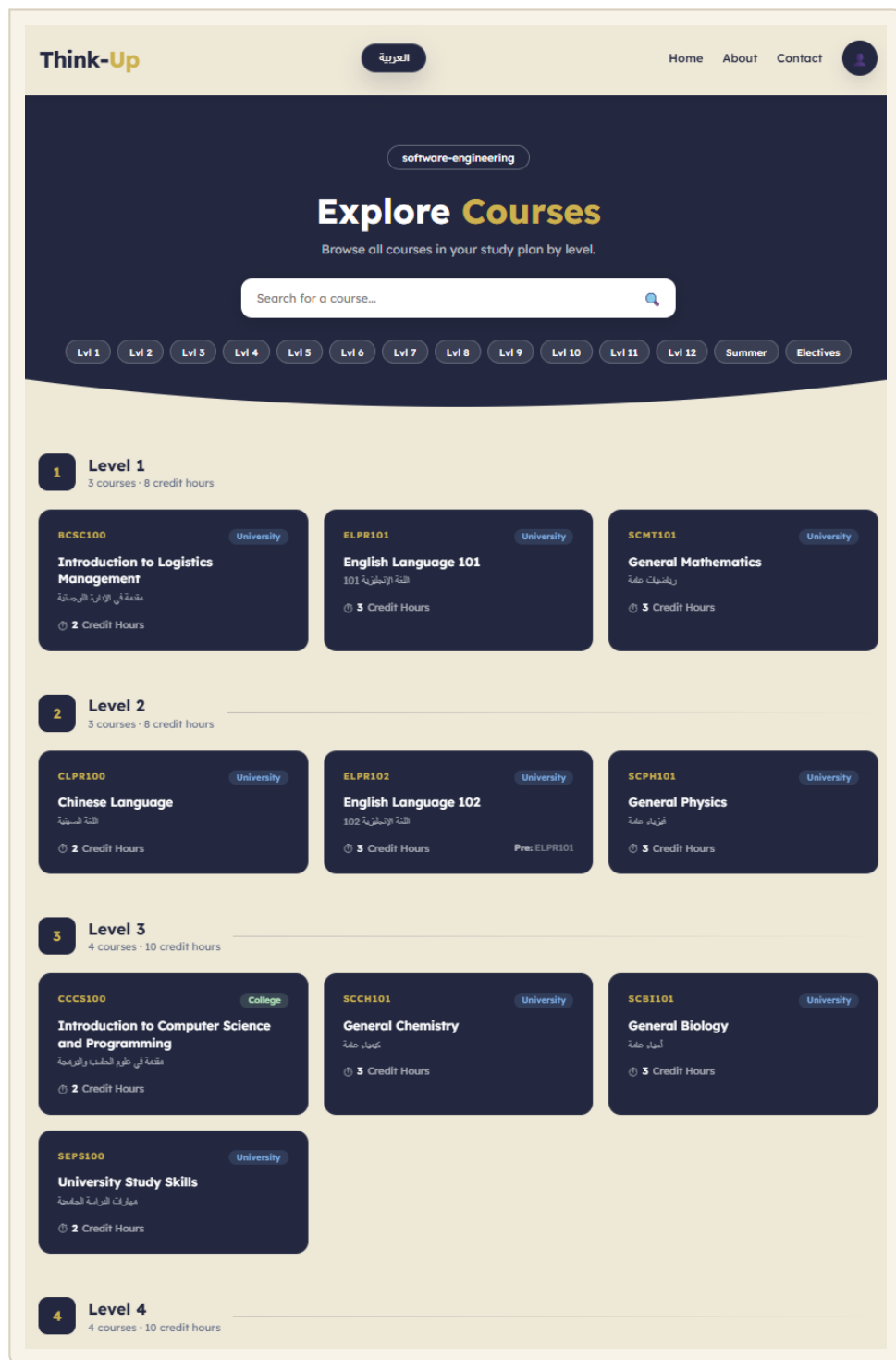


Figure 6 – Step 4 — Explore Courses. Courses are organised by level (1–12, plus Summer and Electives), each card showing its code, bilingual title, and credit hours — drawn live from Firestore.

3.4 Study sessions — the heart of the product

This is where the cross-major idea becomes a live experience. A student opens a course's weekly public schedule, sees which hourly slots are live, open, or full, and either joins an existing session or creates one in an empty slot.

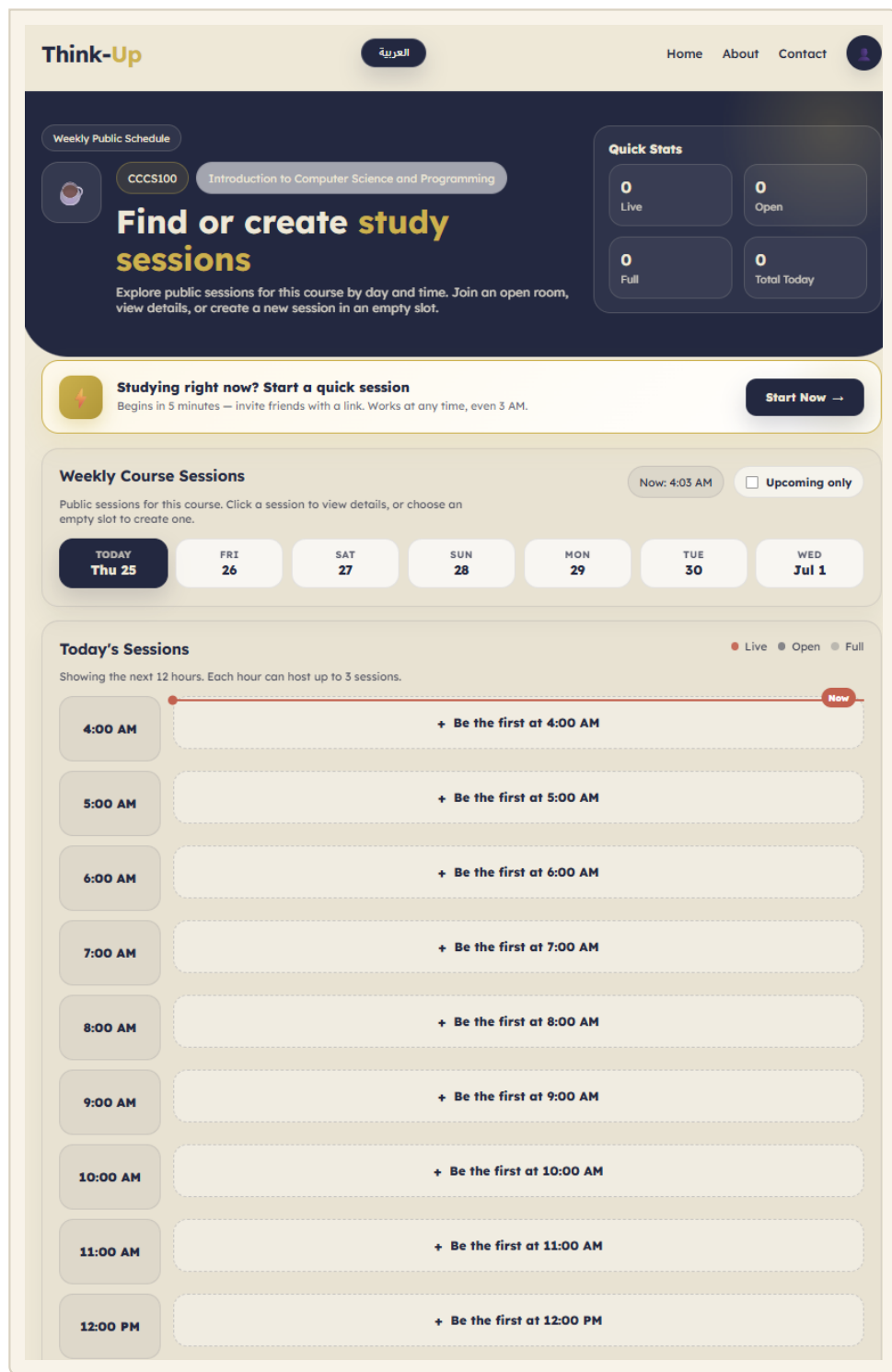


Figure 7 – Find or create sessions. A course's weekly schedule with live quick-stats and an hour-by-hour timeline; any empty slot can be claimed with a single tap.

Think-Up العربية Home About Contact

Session Setup

CCCS100 Introduction to Computer Science and Programming

Create a Study Session

Set your title, time, and visibility. Public sessions appear in the weekly schedule and also generate an invite link you can share.

Session Rules

- 60-minute sessions**
All sessions are 60 minutes — focused and predictable.
- 2 to 6 people**
Smaller groups stay focused. **5 is the sweet spot.**
- Invite link always works**
Even on public sessions, you get a link to share with friends.

Session Details Ready to create

Fill in the required information below to create your new session room.

Session Title
e.g., Final exam review — Chapter 4
A short, clear title for your session (max 60 characters).

Date **Start Time**
06/25/2026 04:00 AM
Choose the session day. Any time of day — even 4 AM if you want.

Duration
60 min
All sessions are one hour

Max Participants **5 people**
2 3 4 5 6
5 is the sweet spot — small enough to stay focused, big enough for diverse perspectives.

Visibility
Public Visible in your major's weekly schedule
Private Hidden from the schedule — invite-only
You'll always get an invite link to share — even for public sessions.

Topic / Notes (Optional)
Examples: Lab 4 review, OOP recap, Midterm practice questions...

☐ I understand that I am responsible for hosting the session respectfully, starting on time, and managing participation.

Create Session **Cancel**

LIVE PREVIEW Public

Introduction to Computer Science and Programming Session

This session will appear in your major's weekly schedule.

DATE	Thursday, Jun 25
TIME	4:00 AM
DURATION	60 min
PARTICIPANTS	5 people
VISIBILITY	Public + invite link

After creating, you'll get an invite link you can share — and your session room will be ready.

Figure 8 – Create a session. A full session-setup form — title, date, time, duration, capacity, and public/private visibility — with a live preview card and the session rules shown side by side. Public sessions appear in the weekly schedule and also generate a shareable invite link.

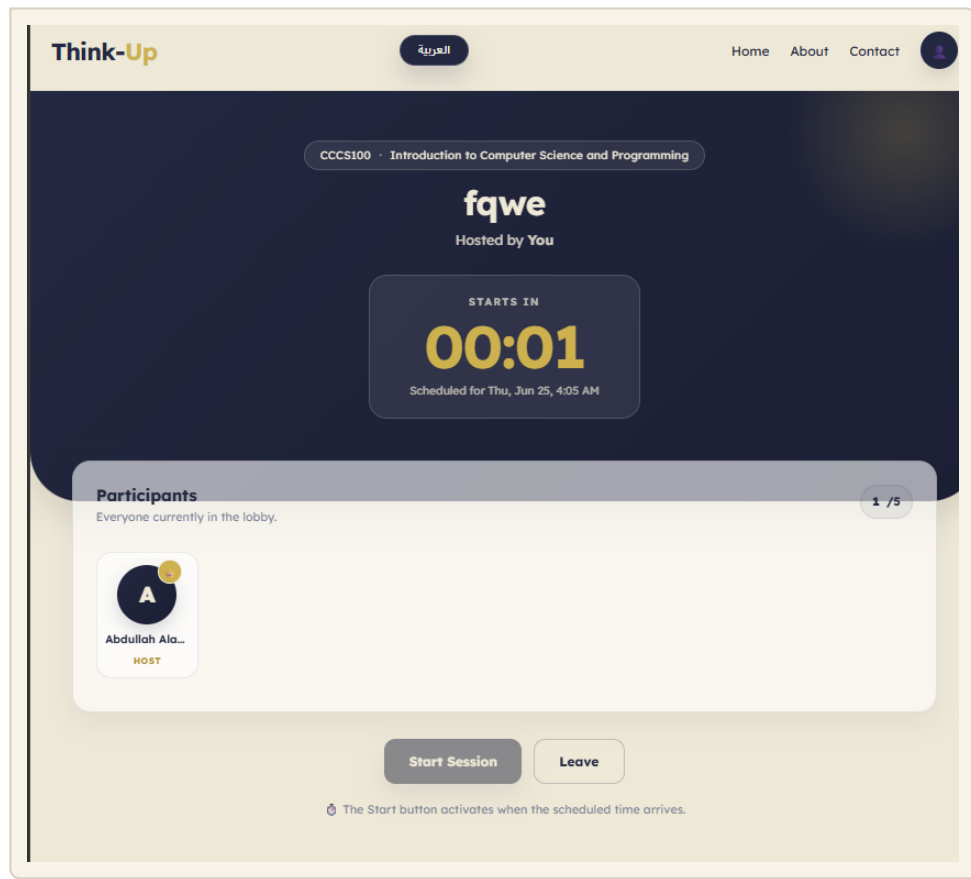


Figure 9 – The session lobby. A real-time waiting room with a live countdown, the participant list, and host controls. The *Start* button activates exactly when the scheduled time arrives — driven by live database listeners, not a page refresh.

3.5 About & Contact

The About page tells the product's story and introduces the team in a structured table; the Contact page is a fully validated, backend-processed enquiry form.



Figure 10 – About — built for students, by students. The product narrative, core values, and a “Meet the builders” table listing each team member with their student ID and links.

Think-Up العربية Home About Contact

Get In Touch

We'd love to hear from you.

Have a question, feedback, or want to partner with Think-Up? Drop us a message and we'll get back to you soon.

Send a message

Fill out the form and our team will respond within 24 hours.

First Name	Last Name
Ahmad	Al-Harbi
Email Address	Mobile Number
you@example.com	05XXXXXXXX
Date of Birth	Gender
mm/dd/yyyy	☐ Male ☐ Female
Preferred Language	Subject
Select language...	Select a topic...
Message	
Write your message here...	
Send Message	

LOCATION

University of Jeddah
College of Computer Science & Engineering
Jeddah, Saudi Arabia

EMAIL

alahmadio313@gmail.com
We reply within 24 hours

SUPPORT HOURS

Sunday - Thursday
9:00 AM - 5:00 PM (AST)

FAQ

Common questions

Is Think-Up free to use? +

Which universities are supported? +

How does the AI summary work? +

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Figure 11 – Contact — a validated enquiry form. Name, email, mobile, date of birth, gender, preferred language, subject, and message — each field validated on the frontend and re-validated and sanitised in a Cloud Function before storage. An FAQ panel answers common questions.

4 Technical Architecture

4.1 Stack at a glance

Frontend	Vanilla HTML5 / CSS3 / JavaScript — no framework, fully hand-authored, responsive to the standard breakpoints.
Database	Cloud Firestore — document model with real-time <code>onSnapshot</code> listeners.
Auth	Firebase Authentication — full sign-up / sign-in lifecycle and access control.
Backend	Firebase Cloud Functions (Node 22, 2nd Gen) for validation, session lifecycle, and integrations.
Live video	Daily.co prebuilt rooms with server-minted access tokens.
Transcription	Deepgram (via Daily) with Arabic language support.
AI layer	Claude API generating structured session summaries.
Secrets	Google Cloud Secret Manager for all third-party keys.

4.2 The data model that makes it work

Three modelling decisions carry most of the product's weight.

Course-anchored sessions

Every session references a `courseCode`, never a `majorId`. This is the relationship that lets students from different majors meet in the same room. It is the architectural foundation of the whole product.

Dual-field participant pattern

Participants are stored two ways at once: a `participantIds` array (efficient for Firestore membership queries) and a `participants` map (constant-time lookup for display and removal), alongside a denormalised `participantCount`. Reads and writes each use the shape that is fastest for them.

Computed session status

A session's state — upcoming, live, ended — is *derived on the client* from its `startTime` and `endTime`, rather than stored as a mutable field that could drift out of sync.

4.3 Real-time behaviour

The lobby, the weekly schedule, and host promotion are all driven by Firestore `onSnapshot` listeners, so every participant sees the same state within moments and without refreshing. Concurrent operations — two students claiming the last seat at once — are made safe with database transactions, and backend operations are written to be idempotent so a retried call never double-applies.

SECURITY MODEL — A DELIBERATE DISTINCTION

Firebase client API keys are public by design and safe to ship in the frontend; they only identify the project. **Third-party** keys — Daily, Claude — are genuine secrets and are kept server-side in Google Cloud Secret Manager, read only by

Cloud Functions. Understanding which keys are sensitive, and why, is a core part of the design rather than an afterthought.

4.4 Live sessions & the AI summary pipeline

A Cloud Function provisions a Daily room and mints a scoped access token per participant; transcription is started automatically at the domain level and the resulting transcript is parsed and written back to Firestore. From there, the transcript feeds the Claude API, which returns a structured summary — `text`, `keyPoints`, `actionItems`, with `status` and `timestamp` — stored as a nested map inside the session document so a student's notes live exactly where the session does.

5 Engineering Challenges & Solutions

Challenge	Resolution
Cross-major collaboration could easily be modelled the wrong way.	Anchored every session to <code>courseCode</code> rather than <code>majorId</code> , making cross-major discovery a property of the schema itself.
Concurrent writes to a single session (last-seat races).	Used Firestore transactions for atomic updates and designed backend operations to be idempotent.
Real-time state without page refreshes.	Built the lobby and schedules on <code>onSnapshot</code> listeners so clients converge automatically.
Keeping live-session config within free-tier and API constraints.	Tuned the Daily token configuration carefully — disabling the prejoin screen, defaulting video off, and starting transcription via the correct token flag.
Protecting third-party credentials.	Centralised Daily and Claude keys in Secret Manager, accessed only by Cloud Functions, while keeping public Firebase keys in the client.
Recovering backend infrastructure after an incident.	Restored all serverless functions and their secret access cleanly from source with a single deploy — a reproducible, infrastructure-as-code workflow.

Design language. The entire interface runs on a tight, deliberate system: **Indigo** (#212842), **Cream** (#F0E7D5), and **Gold** (#D4AF37), set in the **Lexend** typeface — the same palette and font used to typeset this document.

6 Summary

Think-Up began as a course requirement and grew into a working product. Its value does not come from any single feature, but from one structural decision: anchoring study sessions to the *course* a student is taking rather than the *major* they belong to. Everything else — discovery, the weekly schedule, live rooms, AI summaries — is built on top of that one relationship.

Delivering it end to end meant designing a real-time data model, securing a serverless backend, integrating live video with transcription, and wiring an AI summarisation pipeline — while keeping the frontend in hand-written HTML, CSS, and JavaScript and the whole experience bilingual.

IN TOW LINES

A collaborative, AI-assisted study platform that connects Saudi students across majors through the courses they share — designed, built, and shipped end to end.

7 The Team

Think-Up was built by a three-person team, with workload divided across full-stack development, frontend, and integrations.

Abdullah Alahmadi	Full-stack lead — product design, frontend, backend, database, and infrastructure.
Osama Alghamdi	Developer — frontend and feature implementation.
Abdullah AlMazroui	Developer — integrations and supporting features.

Think-Up

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